Unit - 7(Genetics and Evolution) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

Anmol Sharma • Lesson 38 • Feb 16, 2025

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Good evening

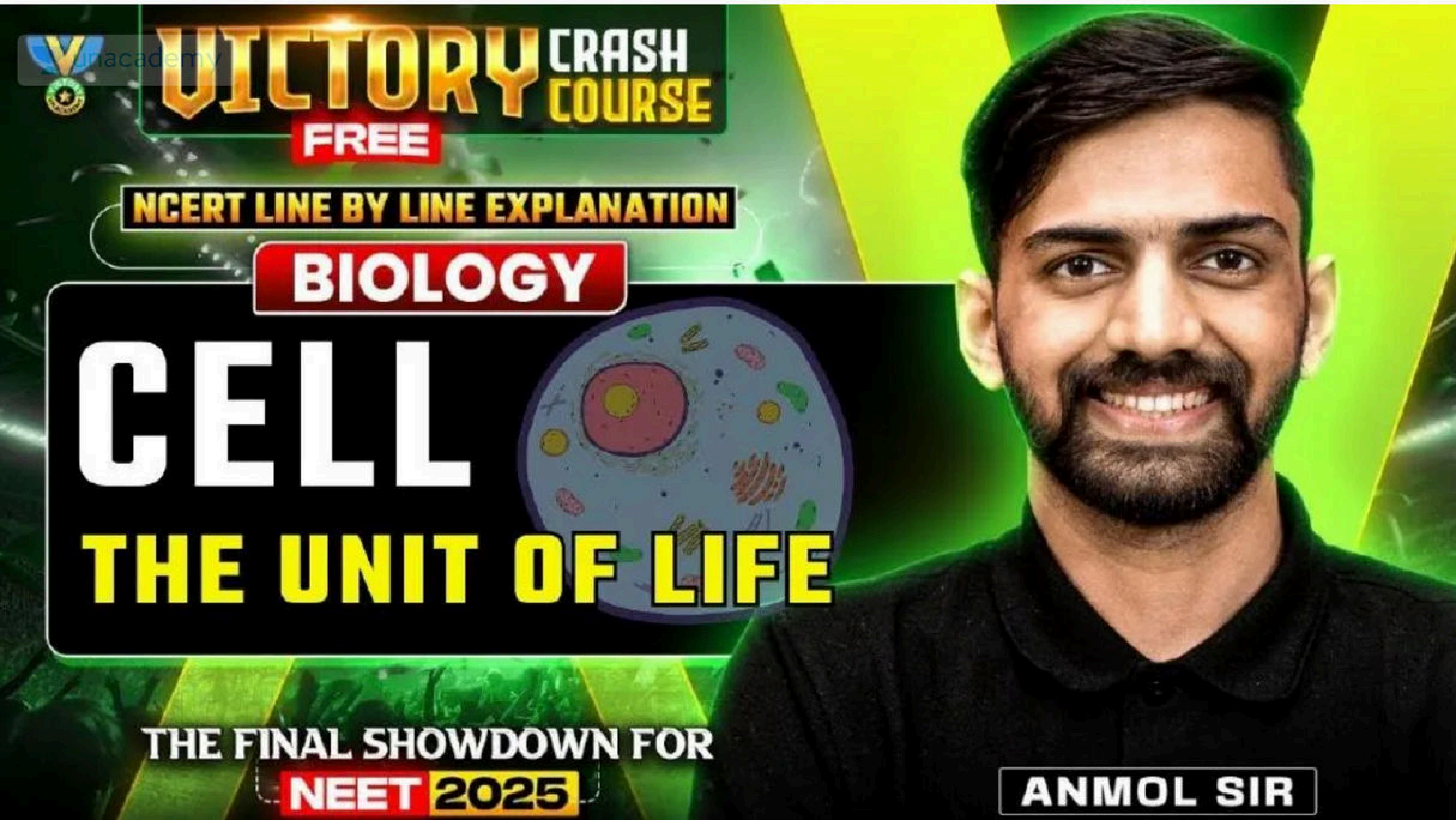
audible ?

Readey ?

200 MCQs

Genetics

BIOLOGYLIVE



VICTORY CRASH COURSE
FREE
NCERT LINE BY LINE EXPLANATION
BIOLOGY
CELL
THE UNIT OF LIFE
THE FINAL SHOWDOWN FOR
NEET 2025
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G.N. Ramachandran
(1922 – 2001)

G.N. RAMACHANDRAN, an outstanding figure in the field of protein structure, was the founder of the 'Madras school' of conformational analysis of biopolymers. His discovery of the triple helical structure of collagen published in *Nature* in 1954 and his analysis of the allowed conformations of proteins through the use of the 'Ramachandran plot' rank among the most outstanding contributions in structural biology. He was born on October 8, 1922, in a small town, not far from Cochin on the southwestern coast of India. His father was a professor of mathematics at a local college and thus had considerable influence in shaping Ramachandran's interest in mathematics. After completing his school years, Ramachandran graduated in 1942 as the top-ranking student in the B.Sc. (Honors) Physics course of the University of Madras. He received a Ph.D. from Cambridge University in 1949. While at Cambridge, Ramachandran met Linus Pauling and was deeply influenced by his publications on models of the α-helix and β-sheet structures that directed his attention to solving the structure of collagen. He passed away at the age of 78, on April 7, 2001.

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When you look around, you see both living and non-living things. You must have wondered and asked yourself – 'what is it that makes an organism living, or what is it that an inanimate thing does not have which a living thing has' ? The answer to this is the presence of the basic unit of life – the cell in all living organisms.

All organisms are composed of cells. Some are composed of a single cell and are called unicellular organisms while others, like us, composed of many cells, are called multicellular organisms.

8.1 What is a Cell?

Unicellular organisms are capable of (i) independent existence and (ii) performing the essential functions of life. Anything less than a complete structure of a cell does not ensure independent living. Hence, cell is the fundamental structural and functional unit of all living organisms.

Anton Von Leeuwenhoek first saw and described a live cell. Robert Brown later discovered the nucleus. The invention of the microscope and its improvement leading to the electron microscope revealed all the structural details of the cell.

1. Which of the following organisms are not composed of cells?

(a) Amoeba
(b) Paramecium
(c) Euglena
(d) None of these

2. Unicellular organisms are not capable of

(a) Independent existence

(b) Performing essential functions of life

(c) Both (a) and (b)

(d) None of these

4 → 9:30

- 10–12
- 6 – 7:30 →
- 9 – 11pm

- 10–11
- $\boxed{1-4}$
- $\boxed{6-7:30}$
- 9 – 11pm → plus

8.2 Cell Theory

In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant. At about the same time, Theodore Schwann (1839), a British Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells. On the basis of this, Schwann proposed the hypothesis that the bodies of animals and plants are composed of cells and products of cells.

Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape. Cell theory as understood today is:

(i) all living organisms are composed of cells and products of cells.

(ii) all cells arise from pre-existing cells.

3. Who was the first one to see a live cell?

(a) Robert Hook
(b) Leeuwenhoek
(c) Robert Brown
(d) None of these

4. Who was the German botanist to study the different cells forming plant tissues?

(a) Schleiden
(b) Schwann
(c) Rudolf Virchow
(d) None of these

5. In which year Matthias Schleiden examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant?

(a) 1638 (b) 1738 (c) 1838 (d) 1938

6. In which year Schwann studied different types of animal cells?

(a) 1839 (b) 1739 (c) 1639 (d) 1938

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1. Which of the following is considered as a recessive character of Pea plant?

(1) Green pod

(2) Round seed

(3) Axial flower

(4) Wrinkled seed → Recessive | Round - Dom.

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2. β Thalassemia is controlled by a single gene :-

(1) HBB → chrom. 11

(2) HBA1

(3) HBA2

(4) HBA1 and HBA2

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3. In the cross Aa Bb X Aa Bb, what proportion of "gametes" are "homozygous" for both the traits

(1) 2/16

(2) 1/4

(3) 0/16

(4) 4/16

AaBb X AaBb

Aa Bb

AB Ab aB ab

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4. In garden pea, starch is synthesised effectively in

BB

- Round

(1) Hybrid round seeded plants

(2) Heterozygous round seeded plants

(3) Wrinkled seeded plants

(4) Homozygous round seeded plants

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5. Broadly, "genetic disorders" may be grouped into two categories –

(1) Morganian disorders & Mendelian disorders

(2) Mendelian disorders and Chromosomal disorders

(3) Morganian disorders and Chromosomal disorders

(4) Morganian disorders and Gene disorders

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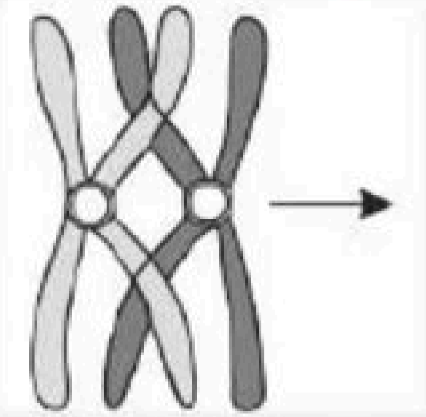
6. The following figure represents a process occurring during phase?



(1) Anaphase - I

(2) Zygotene of Prophase - I

(3) Pachytene of prophase - I

(4) Anaphase - II

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7 The crossing of F_1 hybrid to _____ is done in a test cross is called

		bb	
		b	b
Bb	B	Bb	Bb
	b	bb	bb

(1) F_2

(2) Recessive parent

(3) F_1

(4) Dominant parent

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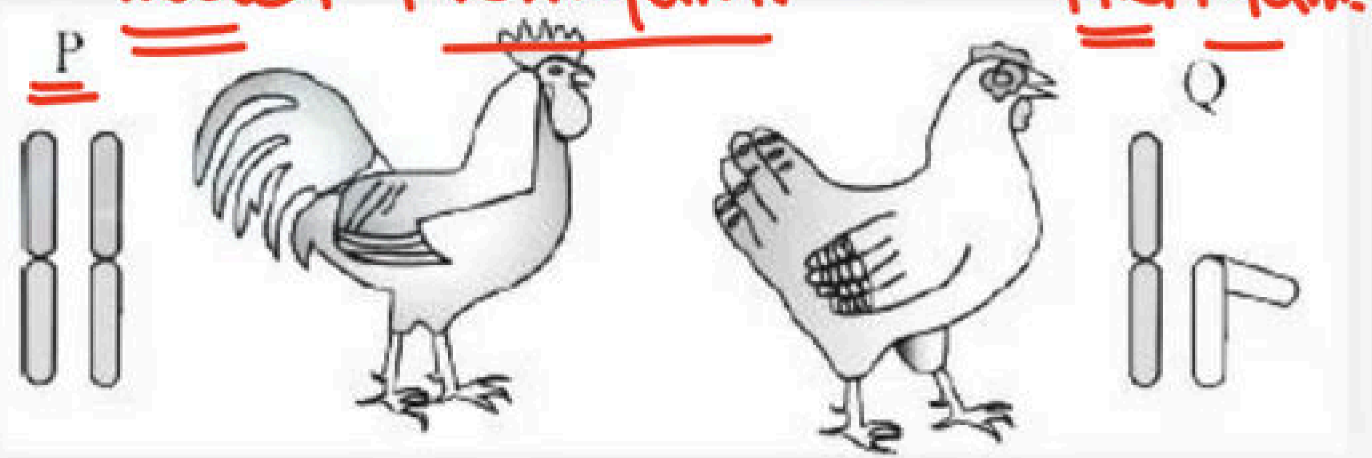
8. Recognise the figure find out the correct matching

male :- Homgam. Hetrgam.



Birds

(1) P = WW Q = ZW

(2) P = XY Q = XX

(3) P = ZZ Q = ZW

(4) P = ZW Q = ZZ

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9. Which statement is true for Law of Segregation

(1) Characters are controlled by discrete units called factors

(2) Factors or alleles of a pair segregate from each other such that a gamete receives only one of the two factors

(3) Mendel crossed the tall plant from F2 with a dwarf plant. This he called a test cross.

(4) Hybrids contain alleles with contrasting traits, they are called homozygous.

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10. ABO blood group is determined by three different alleles. How many genotypes and phenotypes are possible?

(1) Genotype-6 Phenotype-4

(2) Genotype-3 Phenotype-1

(3) Genotype-4 Phenotype-6

(4) Genotype-9 Phenotype-7

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11. Smallest structure having the power of replicating itself is

(1) Ribosome

(2) Gene

(3) Chloroplast

(4) Mitochondria

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12. An individual shows the following characteristics:

A. Sterile females

B. Rudimentary ovaries

C. lack of secondary sexual characters

D. 44 + XO

The individual suffers from :-

(1) Edward's syndrome

(2) Down's syndrome

(3) Turner's syndrome

(4) Klinefelter syndrome

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13. Haemophilia is more common in males because it is a

(1) Recessive character carried by Y chromosome

(2) Dominant trait carried by X chromosome

(3) Recessive trait carried by X chromosome

(4) Dominant character carried by Y chromosome

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14. Aneuploidy is :-

(1) Gain of chromosome

(2) Loss of chromosome

(3) Gain of a gene

(4) Both (1) and (2)

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15. Sickle cell anemia is

(1) Characterized by elongated sickle WBC.

(2) A sex linked dominant trait

(3) Caused by substitution of valine by glutamic acid in the beta globin chain of haemoglobin on chr-11

(4) Caused by a change in a single base pair of DNA

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16. Sickle-cell anaemia is an ____________

(1) Autosome linked dominat trait

(2) Sex- linked dominat trait

(3) Autosome linked recessive trait

(4) Sex- linked recessive trait

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17. Mendel formulated the laws of heredity considering seven pairs of characters in the pea plant. If he had studied an eighth pair, the law which would have been altered is: -

(1) Law of independent assortment

(2) Law of segregation

(3) Law of unit character

(4) Law of dominance

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18. A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind :-

(1) 0.25 ✓

(2) 1

(3) Nil

(4) 0.5

$X^cY \times XX$

$\downarrow$

$X^cX \quad X^cX \quad XY \quad XY$

$\downarrow$

XY

X^cY

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19. Which of the following statements are true?

i. A recessive allele is weaker than a dominant allele X

ii. A recessive allele do not show its effects when paired with a dominant allele ✓

iii. A dominant allele is always better for an organism X

iv. A dominant allele determines the phenotype when paired with a recessive allele ✓

(1) ii, iii and iv

(2) ii and iv ✓

(3) i, ii and iii

(4) i, iii and iv

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20 Statement A : Hybrids are generally back crossed.

Statement B : Test cross is a Back cross .

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

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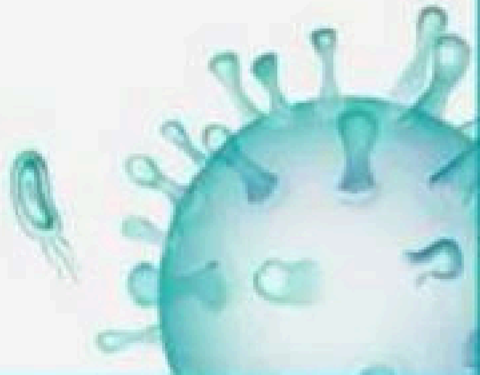
21. Which of the following will not result in variations among siblings?

(1) Independent assortment of genes

(2) Crossing over

(3) Linkage → 2 gene tightly linked = No crossing

(4) Mutation



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22. It is said that Mendel proposed that the factor controlling any character is discrete and independent. This proposition was based on the

(1) Observations that the offspring of a cross made between plants having two contrasting characters shows only one character without any blending

(2) Results of F3 generation of a cross

(3) Cross pollination of parental generations

(4) Self pollination F1 offspring

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23. A human male produces sperms with the genotypes AB, Ab, aB and ab pertaining to two diallelic characters in equal proportions. What is the corresponding genotype of the person ?

(1) AaBb

(2) AaBB

(3) AABB

(4) AABb

AaBb

AB Ab aB ab

AaBB

AB aB

AB

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24. A dihybrid test cross-yielding a result of "1 : 1 : 1 : 1 :" ratio is indicative of:

F1 hy × R.P

$2^2 = 4$

(1) 4 different types of generation dihybrids

(2) 4 different types of gametes produced by the parent

RrYy

(3) 4 different types of gametes produced by the dihybrid

(4) Homozygous condition of the dihybrid

RY Ry rY ry

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25. Red -green colour blindness in human beings is governed by a "sex-linked recessive gene." A normal woman whose father was colourblind marries a colourblind man. What proportion of their sons is expected to be colour blind?

(1) 1/2 ✓ (2) 0/1

(3) 1/4 (4) All

$X^cX \quad X \quad X^cY$

$X^cX^c \quad X^cY \quad X^cX \quad XY$

1

→ 4 = 25%

→ 4 = 50%

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26. Mark the odd one w.r.t syndrome which occur due to "failure of segregation of homologous pair" of chromosomes during cell division cycle.

(1) Turner's syndrome

(2) Down's syndrome

(3) Klinefelter's syndrome

(4) Thalassemia =) mutation or deletion

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27. The technique of removal of stamens in hybridisation used by Mendel was

(1) Emasculation

(2) Back cross

(3) Double cross

(4) Single cross

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28. Match the columns I and II.

(1) A – R, B – P, A – Q

(2) A – Q, B – R, B – P

(3) A – P , B – R, A – Q

(4) A – Q, B – P, B – R

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28. **Match the columns I and II.**

Column I		**Column II**	
A	F_1 Generation	P	3/4 are dominant
B	F_2 Generation	Q	All are dominant, heterozygous
		R	1/4 recessive

(1) A – R, B – P, A – Q

(2) A – Q, B – R, B – P

(3) A – P , B – R, A – Q

(4) A – Q, B – P, B – R

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29. One pair segregates independently of other pair is always applicable for :-

(1) Chromosome

(2) Gene

(3) Neither chromosome nor gene

(4) Both chromosome and gene

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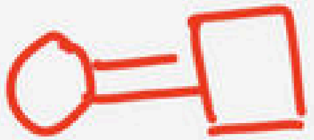
30. Double lines in pedigree analysis show:

(1) Consanguineous marriage

(2) Normal mating

(3) Sex unspecified

(4) Unaffected offspring



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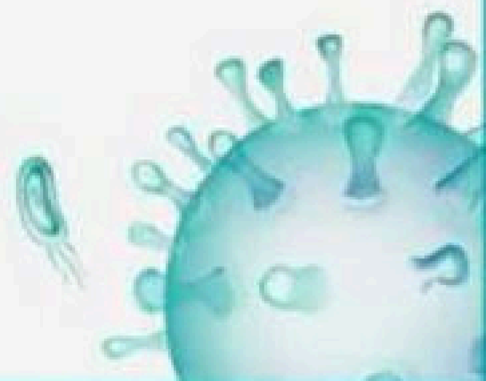
31. Which statements among the following describe Mendels contributions and work?

(1) Mendel published his work on inheritance of characters in 1865

(2) Chromosomal theory of inheritance contained Mendelian principles

(3) Gregor Mendel, conducted hybridisation experiments on garden peas for seven years (1856-1863)

(4) All are correct



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32. Mendel published his work on inheritance of characters in

(1) 1856

(2) 1865 , unrecog. → 1900

(3) 1866

(4) 1863

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33. Select the inappropriate statement :-

A. Morgan observed that the two genes did not segregate independently of each other and the F2 ratio didnot deviated very significantly from the 9:3:3:1

B. When the two genes in a Monohybrid cross were situated on the same chromosome, the proportion of parental gene combinations were much higher than the non-parental type :-

(1) A (2) B

(3) A & B (4) None

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34. In sickle cell anaemia, the sequence of amino acids from the first to the seventh position of the beta -chain of haemoglobins is

(1) His, Leu, Thr, Pro, Glu, Val, Val

(2) Val, His, Leu, Thr, Pro, Glu, Glu

(3) Thr, His, Pro, Val, Pro, Val, Glu

(4) Val, His, Leu, Thr, Pro, Val, Glu

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35. In a cross Drosophila, the heterozygous animal with grey body (b+) and long wings (vg+) was crossed with black body and vestigeal wings, the progeny has the animals in the following ratio : grey vestigeal 24; grey long 126; black long 26; black vestigeal 124.What is the frequency of recombinations in the population?

(1) 16.7%

(2) 15.8%

(3) 14.5%

(4) 17.5%

b^+

$$R.f = \frac{\text{Total no. of Recomb.}}{\text{No. of prog.}}$$

$$= \frac{\cancel{50}}{\cancel{300}_{6}} \times 100$$

$$= \frac{\cancel{100}}{6}$$

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36. An allele is dominant if it is expressed in

(1) Second generation only

(2) Both homozygous and heterozygous states

(3) Homozygous combination only

(4) Heterozygous combination only x

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37. A cross between F1 hybrid and a recessive parent gives a ratio of :

(1) 1 : 1

(2) 3 : 1

(3) 2 : 1

(4) 4 : 1

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38. Mendelian disorders are mainly determined by alteration or mutation in the ________

(1) Single codon

(2) Single base

(3) Multiple gene

(4) Single gene

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39. The phenotype of an individual may be affected if the modified allele produces :-

a. No enzyme at all

b. The normal/less efficient enzyme X

c. A non-functional enzyme

(1) Only (a) is correct

(2) (a) and (c) are correct

(3) (b) and (c) are correct

(4) Only (c) is correct

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40. Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by

(1) Thomas Hunt Morgan

(2) Sutton and Boveri

(3) Alfred Sturtevant

(4) Henking

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41) Among the following characters, which one was not considered by Mendel in his experiments on pea?

2017

(1) Trichomes - Glandular or non-glandular

(2) Seed - Green or Yellow

(3) Pod - Inflated or Constricted

(4) Stem - Tall or Dwarf

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42. A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?

Rr × RR

↓

RR Rr

(1) Only red flowered plants

(2) Red flowered as well as pink flowered plants

(3) Only pink flowered plants

(4) Red, Pink as well as white flowered plants

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43. If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness?

(1) 50%

(2) 75%

(3) 100%

(4) 25%

$X^cX^c \times X^cY$

$X^cX^c \quad X^cY \quad X^cX^c \quad X^cY$

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44. In a test cross involving F1 dihybrid flies, more parental-type offspring were produced than the recombinant type offspring. This indicates

(1) The two genes are linked and present on the same chromosome

(2) The two genes are located on two different chromosomes

(3) Chromosomes failed to separate during meiosis

(4) Both of the characters are controlled by more than one gene

45. Which of the following conditions in human is correctly matched with its chromosomal abnormality/ linkage?

(1) Klienefelter's syndrome - 44 autosomes + XXY

(2) Down's syndrome - 44 autosomes + XO X

(3) Erythroblastosisfoetalis - X linked

(4) Colour blindness - Y linked

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46 A pleiotropic gene

(1) Controls a trait only in combination with another gene

(2) Is expressed only in primitive plants

(3) Controls multiple traits in an individual

(4) Is a gene evolved during Pliocene.

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47. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits?

(1) 2

(2) 14

(3) 8

(4) 4

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48. In Antirrhinum (Snapdragon), a red flower was crossed with a white flower and in F1 generation pink flowers were obtained. When pink flowers were selfed, the F2 generation showed white, red and pink flowers.

Choose the incorrect statement from the following :

(1) This experiment does not follow the Principle of Dominance.

(2) Pink colour in F_1 is due to incomplete dominance.

(3) Ratio of F_2 is $\frac{1}{4}$ (Red) : $\frac{2}{4}$ (Pink) : $\frac{1}{4}$ (White)

(4) Law of Segregation does not apply in this X

universal law

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49. If two persons with 'AB' blood group marry and have sufficiently large number of children could be classified as 'A' blood group : 'AB' blood group: 'B' blood group in 1:2:1 ratio. Modern technique of protein electrophoresis reveals presence of both 'A' and 'B' type proteins in 'AB' blood group individuals. This is an example of

(1) Incomplete dominance

(2) Co-dominance

(3) Complete dominance

(4) Partial dominance

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50. Select the correct match.

(1) Phenylketonuria-Autosomal dominant trait

(2) Sickle cell anaemia-Autosomal recessive trait, chromosome-11

(3) Thalassemia- X linked

(4) Haemophilia-Y linked

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51. Multiple alleles are present:

(1) On different chromosomes

(2) On non-sister chromatids

(3) At the same locus on the chromosome

(4) At different loci on the same chromosome

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52. Which one from those given below is the period for Mendel's hybridization experiments?

(1) 1840-1850

(2) 1857-1869

(3) 1870-1877

(4) 1856-1863

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53. A disease caused by an autosomal primary nondisjunction is :

(1) Klinefelter's Syndrome

(2) Turner's Syndrome

(3) Sickle Cell Anemia => point mut

(4) Down's Syndrome

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54. Phenotype of an organism is the result of

(1) Cytoplasmic effects and nutrition

(2) Mutations and linkages

(3) Genotype and environment interactions

(4) Environmental changes and sexual dimorphism

55. Select the correct statement from the ones given below with respect to dihybrid cross :-

(1) Genes far apart on the same chromosome show very few recombination's

(2) genes loosely linked on the same chromosome show similar recombination's as the tightly linked ones

(3) Tightly linked genes on the same chromosomes show very few recombination's

(4) Tightly linked genes on the same chromosomes show higher recombination's

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56. What is the genetic disorder in which an individual has an overall masculine development gynaecomastia, and is sterile ?

(1) Turner's syndrome

(2) Klinefelter's syndrome

(3) Edward syndrome

(4) Down's syndrome

57. In his classic experiments on pea plants, Mendel did not used :-

(1) Seed shape

(2) Seed colour

(3) Flower position

(4) Podlength

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58. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased?

(1) 50%

(2) 75%

(3) 25%

(4) 100%

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59. XO type of sex determination can be found in

(1) Birds

(2) Grasshoppers

(3) Monkeys

(4) Drosophila

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60. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

In the light of the above statements, choose the correct answer from the options given below:

Assertion : Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.

Reason : Closely located genes assort independently.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

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61. Select the incorrect statement.

(1) Male fruit fly is heterogametic

(2) In male grasshoppers 50% of sperms have no sexchromosome

(3) In domesticated fowls, sex of progeny depends on the

ovum

type of ~~sperm~~ rather than egg

(4) Human males have one of their sexchromosome

much shorter than the other

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62. The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosomes?

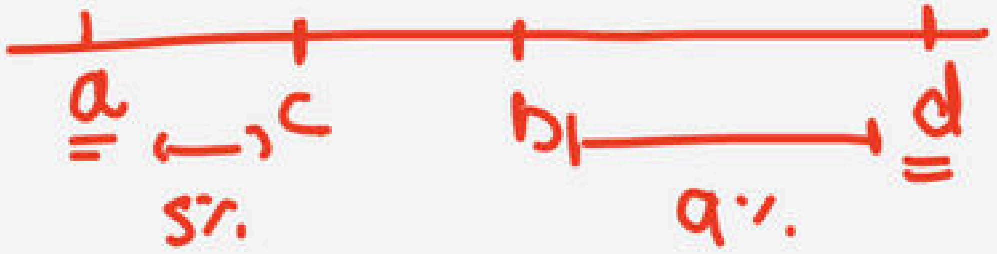
(1) d, b, a, c ✗

(2) a, b, c, d

(3) a, c, b, d ✓

(4) a, d, b, c ✗

- a and c = 5%
- b and c = 15%
- b and d = 9%
- a and b = 20%
- c and d = 24%
- a and d = 29%



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63. Experimental verification of the chromosomal theory of inheritance was done by-

(1) Sutton

(2) Boveri

(3) Morgan

(4) Mendel

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64. Match Column I with Column II

Column I		Column II	
A	Two or more alternative forms of a gene	P	Back cross
B	Cross of F_1 progeny with homozygous recessive parent	Q	Ploidy
C	Cross of F_1 progeny with any of the parents	R	Allele
D	Number of chromosome sets in plant	S	Test cross

(1) A – P, B – Q, C – R, D – S
(2) A – Q, B – P, C – R, D – S
(3) A – R, B – S, C – P, D – Q
(4) A – S, B – R, C – Q, D – P

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65. In a plant, black seed color is dominant over white seed color (BB/bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

BB x bb

(Bb, BB) x bb

(1) BB

✓(2) bb

(3) Bb

(4) BB / Bb

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66. Which of the following occurs due to the presence of autosome linked dominant trait?

(1) Myotonic dystrophy

(2) Haemophilia

(3) Thalessemia

(4) Sickle cell anaemia

BIOLOGYLIVE

Title: 200 MCQs Genetics

67. An abnormal human baby with 'XXX' sex chromosomes was born due to

(1) Formation of abnormal sperms in the father

(2) Fusion of two sperms and one ovum

(3) Fusion of two ova and one sperm

(4) Formation of abnormal ova in the mother

unacademy

68. Genetic variation in a population arises due to

(1) Reproductive isolation and selection

(2) Recombination only

(3) Mutations as well as recombination

(4) Mutations only

BIOLOGYLIVE

unacademy

69. Which of the following pairs is wrongly matched?

(1) XO type sex : Grasshopper determination

(2) ABO blood grouping : Co-dominance

(3) Starch synthesis in pea : ~~Multiple alleles~~

Pleiotrop

(4) T.H. Morgan : Linkage

BIOLOGYLIVE

unacademy

70. Broad palm with single palm crease is visible in a person suffering from-

(1) Down's syndrome

(2) Turner's syndrome

(3) Klinefelter's syndrome

(4) Thalassemia

BIOLOGYLIVE

unacademy

71. A cross in which an organism showing a dominant phenotype is crossed with the recessive parent in order to know its genotype is called

(1) Test cross

(2) Monohybrid cross

(3) Back cross

(4) Dihybrid cross

BIOLOGYLIVE

unacademy

72. Given below are two statements :

Statement A : Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance

Statement B : Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

BIOLOGYLIVE

unacademy

73. Thalassemia and sickle cell anemia are caused due to a problem in globin molecule synthesis. Select the correct statement :

(1) Both are due to a quantitative defect in globin chain synthesis

(2) Thalassemia is due to less synthesis of globin molecules

(3) Both are due to a qualitative defect in globin chain molecules

(4) Both are due to a qualitative defect in globin chain synthesis

BIOLOGYLIVE

Title: 200 MCQs Genetics

74. Match Column I with Column II:

Column I		Column II	
A	Down's syndrome	P	11^{th} chromosome
B	α-Thalassemia	Q	'X' chromosome
C	β-Thalassemia	R	21^{th} chromosome
D	Klinefelter's syndrome	S	16^{th} chromosome

(1) $A-P, B-Q, C-R, D-S$

(2) $A-Q, B-R, C-S, D-P$

(3) $A-R, B-S, C-P, D-Q$

(4) $A-S, B-P, C-Q, D-R$

Title: 200 MCQs Genetics

75. Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles do not show any expression and both the characters appear as such in F_2 generation. ✗

C. Factors occur in pairs in normal diploid plants. ✓

D. The discrete unit controlling a particular character is called factor. ✓

E. The expression of only one of the parental characters is found in a monohybrid cross. ✓

Choose the correct answer from the options given below:

(1) A, B and C only

(2) A, C, D and E only

(3) B, C and D only

(4) A, B, C, D and E

unacademy

76. A woman has an X-linked condition on one of her X chromosomes. This chromosome can be inherited by

(1) Only grandchildren

(2) Only sons

(3) Only daughters

(4) Both sons and daughters

BIOLOGYLIVE

unacademy

77. F_2 generation in a Mendelian cross showed that both genotypic and phenotypic ratios are same as 1:2:1. It represents a case of :

(1) Dihybrid cross

(2) Multiple allelism

(3) Monohybrid cross with complete dominance

(4) Monohybrid cross with incomplete dominance

BIOLOGYLIVE

unacademy

Title: 200 MCQs Genetics

78. As per ABO blood grouping system, the blood group of father is B+, mother is A+ and child is O+. Their respective genotype can be

A. $I^{Bi}/I^{Ai}/ii$
B. $I^{B}I^{B}/I^{A}I^{A}/ii$
C. $I^{A}I^{B}/iI^{A}/I^{B}$
D. $I^{A}i\ /\ I^{B}i\ /\ I^{A}i$
E. $iI^{B}/iI^{A}/I^{A}I^{B}$

$I^{A}i \times I^{B}i$

$I^{A}I^{B}$ ii

Choose the most appropriate answer from the options given below :

(1) A only
(2) B only
(3) C & B only
(4) D & E only

BIOLOGYLIVE

unacademy

79. Which of the following characteristics represent 'Inheritance of blood groups' in humans?

A. Dominance

B. Co-dominance

C. Multiple allele

D. Incomplete dominance

E. Polygenic inheritance

(1) B, D and E

(2) A, B and C

(3) B, C and E

(4) A, C and E

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80. The production of gametes by the parents, formation of zygotes, the F1 and F2 plants, can be understood from a diagram called:

(1) Bullet square

(2) Punch square

(3) Punnett square

(4) Net square

BIOLOGYLIVE

unacademy

81. Which of the following statements are correct about Klinefelter's Syndrome?

A. This disorder was first described by Langdon Down (1866). x

B. Such an individual has overall masculine development. However, the feminine developement is also expressed.

C. The affected individual is short statured. x

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile. x

Choose the correct answer from the options given below:

(1) A and B only
(2) C and D only
(3) B and E only
(4) A and E only

BIOLOGYLIVE

unacademy

82. The genotypes of a husband and wife are IAIB and IAi. Among the blood types of their children, how many different genotypes and phenotypes are possible?

(1) 3 genotypes ; 4 phenotypes

(2) 4 genotypes ; 3 phenotypes

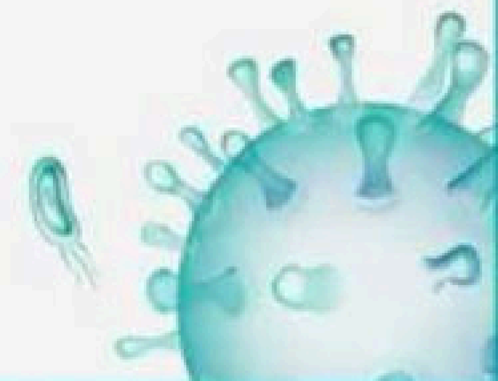
(3) 4 genotypes ; 4 phenotypes

(4) 3 genotypes ; 3 phenotypes

$I^A I^B \times I^A i$

$I^A I^A \quad I^A i \quad I^B I^A \quad I^B i$

A A AB B



BIOLOGYLIVE

Title: 200 MCQs Genetics

83. Distance between the genes and percentage of recombination shows:

(1) A direct relationship

(2) An inverse relationship

(3) A parallel relationship

(4) No relationship

$\uparrow\downarrow$ C.O $\propto \dfrac{1}{\text{link}\uparrow\downarrow}$

unacademy

84. Occasionally, a single gene may express more than one effect. The phenomenon is called:

(1) Multiple allelism

(2) Mosaicism

(3) Pleiotropy

(4) Polygeny

BIOLOGYLIVE

unacademy

85. Conditions of a karyotype 2n +1, 2n –1 and 2n + 2, 2n – 2 are called:

(1) Aneuploidy

(2) Polyploidy

(3) Allopolyploidy

(4) Monosomy

Title: 200 MCQs Genetics

86. Z Z / ZW type of sex determination is seen in:

(1) Platypus

(2) Snails

(3) Cockroach

(4) Peacock

BIOLOGYLIVE

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Title: 200 MCQs Genetics

87. Mother and father of a person with 'O' blood group have 'A' and 'B' blood group, respectively. What would be the genotype of both mother and father?

(1) Mother is homozygous for 'A' blood group and father is heterozygous for 'B'

(2) Mother is heterozygous for 'A' blood group and father is homozygous for 'B'

$I^B i$ $I^A i$

(3) Both mother and father are heterozygous for 'A' and 'B' blood group, respectively

(4) Both mother and father are homozygous for 'A' and 'B' blood group, respectively

BIOLOGYLIVE

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88. Two genes 'A' and 'B' are linked. In a dihybrid cross involving these two genes, the F1 heterozygote is crossed with homozygous recessive parental type (aa bb). What would be the ratio of offspring in the next generation?

(1) 1 : 1 : 1 : 1 → unlinked

(2) 9 : 3 : 3 : 1

(3) 3 : 1

(4) 1 : 1

BIOLOGYLIVE

89. Match the columns I (Characters) and II. (Contrasting Traits)

Column I	Column II
A Flower colour	P White
B Pod shape	Q Yellow
C Pod colour	R Constricted
D Seed shape	S Wrinkled

(1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P , B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

90. Match the columns I and II.

Column I		Column II	
A	Hemophilia	P	Absence of particular pigments in cones
B	Colour blindness	Q	Clotting protein
C	Pedigree analysis	R	Identify inheritance in families
D	Sickle cell anaemia	S	Mutation

(1) A – R, B – P , C – Q, D – S

(2) A – Q, B – R, C – P, D – S

(3) A – P , B – R, C – Q, D – S

(4) A – Q, B – P , C – R, D – S

91. Match the columns I and II.

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. Imp

Column I		Column II	
A	Monohybrid phenotype	P	3:1
B	Monohybrid genotype	Q	1:2:1:2:4:2:1: 2:1
C	Dihybrid genotype	R	1:2:1
D	Dihybrid phenotype	S	9:3:3:1

(1) A – R , B – P , C – Q, D – S

(2) A – Q , B – R , C – P, D – S

(3) A – P , B – R , C – Q, D – S

(4) A – Q , B – P , C – R , D – S

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92. Match the columns I and II with respect to Colour blindness

Column I		Column II	
A	Normal female	P	X^cX^c
B	Carrier female	Q	X^CX^C
C	Diseased female	R	X^CX^c
D	Diseased male	S	X^cY^c

XX

(1) A – R , B – P , C – Q, D – S

(2) A – Q , B – R , C – P, D – S

(3) A – P , B – R , C – Q, D – S

(4) A – Q , B – P , C – R , D – S

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93. Match Column I with Column II and find the correct answer

Column I		Column II	
A	Aneuploidy	P	Increase in whole set of chromosomes
B	Monoploidy	Q	Loss or gain of a chromosome
C	Polyploidy	R	Two sets of chromosomes
D	Diploidy	S	A single set of chromosomes

(1) A - P, B - Q, C - R, D – S

(2) A - R, B - P, C - Q, D - S

(3) A - Q, B - S, C - P, D – R

(4) A - S, B - R, C - P, D - Q

BIOLOGYLIVE

94. Match the columns I and II.

Column I		Column II	
A	Hemophilia	P	Recessive chromosome 12
B	Phenylketonuria	Q	Recessive X chromosome anaemia
C	Thalassemia A	R	Chromosome 16
D	Sickle cell anaemia	S	Autosomal chromosome 11

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

(3) A – P , B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

4

Title: 200 MCQs Genetics

95. Match the columns I and II to Hemophilia.

Column I		Column II	
A	Normal female	P	X^hX^h
B	Carrier female	Q	X^HX^H
C	Diseased female	R	X^HX^h
D	Diseased male	S	X^hY

(1) A – R , B – P , C – Q, D – S

(2) A – Q , B – R , C – P, D – S

(3) A – P , B – R , C – Q, D – S

(4) A – Q , B – P , C – R , D – S

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96. Match the columns I and II w.r.t. Sex determination. Column I Column II

Column I	Column II
A Haplo-diploid system	P Female heterogamety
B ZW - ZZ method	Q Male heterogamety
C XX - XY method	R Apis
D XX - XO method	S Locust

(1) A – R , B – P , C – Q, D – S

(2) A – Q , B – R , C – P, D – S

(3) A – P , B – R , C – Q, D – S

(4) A – Q , B – P , C – R , D – S

BIOLOGYLIVE

97. Match the columns I and II

Column I	Column II
A Phenylketonuria	P X-chromosome
B Sickle cell anaemia	Q Chr.12
C Hemophilia	R Chr.11

(1) A – R , B – P , C – Q

(2) A – Q , B – R , C – P

(3) A – P , B – R , C – Q

(4) A – Q , B – P , C – R

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98. Select the correct statement with respect to pedigree analysis

i. Control crosses can be performed in pea plant or some other organisms.

ii. Control crosses are not possible in case of human beings.

iii. Study of the family history about inheritance of a particular trait provides an alternative.

iv. The inheritance of a particular trait is represented in the family ~~garden~~ tree over generations.

(1) All are correct except i

(2) All are correct except ii

(3) All are correct except iii

(4) All are correct except iv

BIOLOGYLIVE

unacademy

99. Statement A : In birds the sex of the young ones is determined by the sex chromosomes of the mother.

Statement B : Mother has ZW sex chromosomes

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.

BIOLOGYLIVE

unacademy

100. Select the incorrect statement with respect to Sicklecell anaemia

8am = 'uua

a. It is an example of lethal gene ✓

b. It involves the 11th chromosome ✓

c. All the carriers die when they are born ×

d. Under low oxygen tension, RBCs break due to sickle formation

(1) Only a ✓(2) Only c

(3) Only b (4) Only d

BIOLOGYLIVE

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(उच्चतर शिक्षा विभाग, शिक्षा मंत्रालय, भारत सरकार के तहत एक स्वायत्त संगठन)
(An Autonomous Organization under the Department of Higher Education, Ministry of Education, Government of India)

PUBLIC NOTICE

25th January 2025

Subject:Clarification on Question Paper Pattern and Examination Duration for NEET (UG)-2025

The National Testing Agency has been conducting the NEET (UG) since 2019 on behalf of National Medical Commission.

All NEET (UG)-2025 aspirants are hereby informed that the question paper pattern and examination duration will revert to the pre-COVID format, where there will not be any Section B anymore. Hence, there will be a total of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 in Biology) which will be attempted by the candidates in 180 minutes thereby removing any optional questions and extra time introduced due to COVID.

900 → Type of Q

Question Paper Pattern:
The question paper will consist of 180 compulsory questions.

The provision for optional questions, introduced temporarily during the COVID-19 pandemic , will no longer be available.

Examination Duration:
The total duration of the examination will be 180 minutes (3 hours).

Candidates are advised to prepare accordingly and regularly check the official NTA website https://neet.nta.nic.in for further updates. For further clarification related to NEET (UG)–2025 , contact 011-40759000 or email at neetug2025@nta.ac.in .

Director Exam

NIOS

Solved MCQs →

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Sr. no	Chapter Name	Date
*7days 1	Cell: The Unit life	17th Feb 2025
2	"Cell Cycle and Cell Division"	18th Feb 2025
3	Principle of Inheritance and Variation	19th Feb 2025
4	Molecular Basis of Inheritance	20th Feb 2025
5	Evolution	21st Feb 2025
6	Biotecnology : Principle and process	22nd Feb 2025
7	Biotecnology : Application	24th Feb 2025

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101. There are 64 types of codons in genetic code dictionary because

(1) Genetic code is triplet

(2) There are 64 types of tRNAs found in cell

(3) There are 44 meaningless and 20 codons for amino acids

(4) There are 64 amino acids for coding

BIOLOGYLIVE

102. The Okazaki fragments in DNA chain growth

(1) Prove semiconservative nature of DNA replication

(2) Polymerize in the 3' to 5' direction and forms replication fork

(3) Polymerize in the 5' to 3' direction and explain 3' to 5' DNA replication

(4) Result in transcription

103. Match the columns I and II.

Column I		Column II	
A	DNA-dependent DNA polymerase	P	Replication is continuous
B	Deoxyribonucleoside triphosphates	Q	Average rate of polymerisation 2000 bp per second
C	Polarity $3' \rightarrow 5'$ of parental strand	R	Dual prupose
D	Polarity $5' \rightarrow 3'$ of oarental strand	S	Replication is discontinuous

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

Title: 200 MCQs Genetics

104. RNA primer helps in :

(1) Transcription

(2) Translation

(3) Replication

(4) Transformation

BIOLOGYLIVE

unacademy

105. t-RNA for serine and tyrosine bears anticodon_____.

(1) AGU, AUG

(2) UAC, AUG

(3) UCA, AUG

(4) UGA, AUG

BIOLOGYLIVE

106. Amino acid sequence, in protein synthesis is decided by the sequence of

(1) mRNA

(2) cDNA

(3) rRNA

(4) tRNA

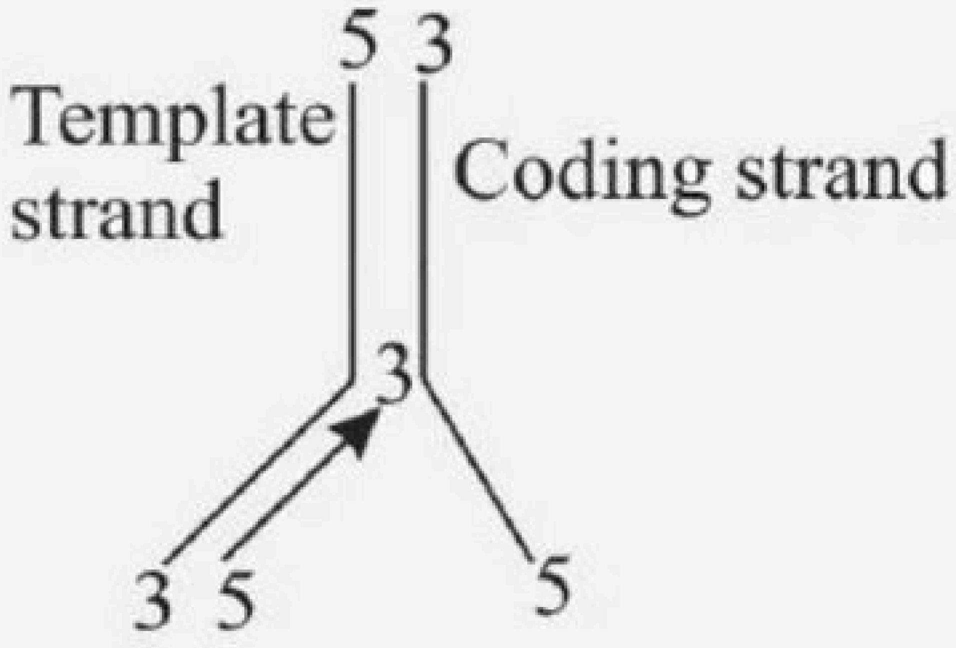
107. This Diagram indicates

(1) DNA replication

(2) Reverse transcription

(3) Transcription

(4) Translation



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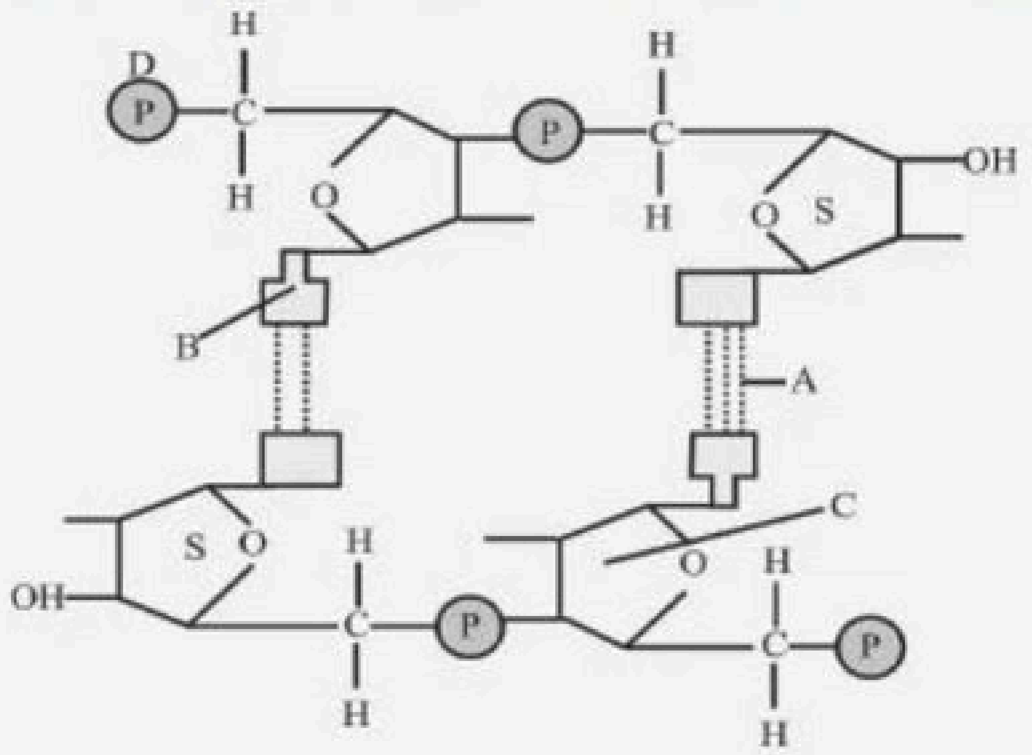
108. Study the given portion of double stranded polynucleotide chain carefully. Identify A, B, and D.



(1) A-phosphodiester bond, B-purine, C-phosphate, Dpentose sugar

(2) A-hydrogen bonds, B-pyrimidine, C-pentose sugar, Dphosphate

(3) A-phosphodiester bond, B-pyrimidine, C-phosphate, Dpentose sugar

(4) A-hydrogen bonds, B-purine, C-pentose sugar, Dphosphate

BIOLOGYLIVE

unacademy

109. Removal of introns and joining the exons in a defined order in a transcription unit is called

(1) Splicing

(2) Tailing

(3) Transformation

(4) Capping

BIOLOGYLIVE

110. The one aspect which is not a salient feature of genetic code, is its being

(1) Ambiguous

(2) Specific

(3) Degenerate

(4) Universal

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111. Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology?

(1) Genetic code is not ambiguous

(2) Genetic code is redundant

(3) Genetic code is nearly universal

(4) Genetic code is specific

BIOLOGYLIVE

112. Which enzyme will be produced in a cell in if there is a nonsense mutation in the lac Y gene?

(1) -galactosidase

(2) Transacetylase

(3) Lactose permease and transacetylase

(4) Lactose

unacademy

113. Which one of the following pairs of codons is correctly matched with their function or the sign for the particular amino acid?

(1) GUU, GCU - Alanine

(2) AUG, ACG - Start / Methionine

(3) UUA, UCA - Leucine

(4) UAG, UGA - Stop

BIOLOGYLIVE

114. Identify the correct statement.

(1) In capping, methyl guanosine triphosphate is added to the 3 ' end of hnRNA.

(2) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

(3) The coding strand in a transcription unit is copies to an mRNA.

(4) Split gene arrangement is characteristic of prokaryotes.

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115. In contrast to DNA Polymerase, RNA Primase

(1) Synthesis of RNA primer that initiates DNA synthesis

(2) Fills the gap between Okazaki fragment

(3) Catalyses new strand in 5′ → 3′ direction

(4) Checks & edits simultaneously during synthesis

BIOLOGYLIVE

unacademy

116. Find the incorrect match.

(1) $\phi \times 174$ phage -5368 deoxyribose sugar in its genetic material

(2) Haploid content of human DNA is 6.6×10^9 bp

(3) E.coli - 4.6×10^6 base pairs

(4) Both 1 and 2

BIOLOGYLIVE

unacademy

117. Assertion : Transcription is the mode in which DNA passes its genetic information to RNA

Reason : Transcription takes place in the cytoplasm of eukaryotic cells.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect

BIOLOGYLIVE

unacademy

118. Each individual has a unique DNA Fingerprint as individuals relies on

(1) Location of minisatellites on chromosome

(2) Number of minisatellites on chromosome

(3) Size of minisatellites on chromosome

(4) All the above

BIOLOGYLIVE

Title: 200 MCQs Genetics

119. Match the columns I and II

Column I	Column II
A i gene	P Transacetylase
B z gene	Q Repressor of the lac operon
C y gene	R Permease
D a gene	S Betagalactosidase

(1) A – R,B – P,C – Q,D – S
(2) A – Q,B – R,C – P,D – S
(3) A – P,B – R,C – Q,D – S
(4) A – Q,B – S,C – R,D – P

120. Read statement 1 - 4

1. In transcription thymine pairs with uracil

2. Regulation of lac operon by repressor is positive and negative regulation

3. Human genome has appoximate 80,000 genes

4. One nucleosome contain 146 bp

How many of the above statements are correct

(1) 2 (2) 3

(3) 4 (4) 1

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121. HGP- the largest known human gene was of:

(1) Dystrophin found on X chromosome

(2) Dystrophin found on Y chromosome

(3) Dystrophin found on 1 chromosome

(4) Dystrophin found on X/Y chromosome

BIOLOGYLIVE

unacademy

122. What is incorrect for human chromosome 1 ?

(1) It is one of the largest chromosome

(2) Its sequence was completed in May 2006

(3) Chromosome 1 has most genes (2968)

(4) It was the first chromosome to be sequenced

BIOLOGYLIVE

123. Assertion : DNA is a better genetic material.

Reason : DNA chemically is less reactive and structurally more stable when compared to RNA.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect

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124. VNTR varies in size from

(1) 0.1 – 20 mb

(2) 0.1 – 20 kb

(3) 0.1 – 20 gb

(4) 0.4 – 15 kb

BIOLOGYLIVE

unacademy

125. ELSI strands for:

(1) Ecological, Logical and Society Issues

(2) Ethical, Legal and Social Issues

(3) Ethical, Logical and Social Issues

(4) Economical, Legal and Society Issues

BIOLOGYLIVE

unacademy

126. In DNA percentage of thymine is 20 , then what is the percentage of guanine?

(1) 20

(2) 40

(3) 30

(4) 60

BIOLOGYLIVE

unacademy

127. Most of the amino acids are represented by more than one codons, the genetic code is said to be

(1) Degenerate

(2) Deaminated

(3) Comma less

(4) Overlapping

BIOLOGYLIVE

unacademy

128. Select the correct option

A. Introns are said to be those sequence that appear in mature or processed RNA

B. Exons or intervening sequences do not appear in mature or processed RNA.

C. The split-gene arrangement further complicates the definition of a gene in terms of a DNA segment

D. Inheritance of a character is also affected by promoter and regulatory sequences of a structural gene.

(1) A and B are correct
(2) C and B are correct
(3) B and D are correct
(4) C and D are correct

BIOLOGYLIVE

unacademy

129. Why glucose cannot act as an inducer for lac operon?

(1) Because they cannot bind with the repressor

(2) Because they cannot bind with the operator

(3) Because they cannot bind with the promoter

(4) Because they cannot bind with the lac operon

BIOLOGYLIVE

Title: 200 MCQs Genetics

130. Formation of a Peptide bond requires energy. Therefore, amino acids are activated in the presence_____ of and linked to their cognate tRNA - a process commonly called as charging of tRNA or____ to be more specific.

(1) GTP, aminoacylation of tRNA

(2) ATP, aminoacylation of mRNA

(3) ATP, amination of tRNA

(4) ATP, aminoacylation of tRNA

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131. Which enzyme is responsible for the synthesis of tRNA in prokaryotes ?

(1) DNA-dependent RNA polymerase

(2) RNA-dependent RNA polymerase

(3) DNA-dependent DNA polymerase

(4) RNA-dependent DNA polymerases

BIOLOGYLIVE

unacademy

132. Statement A : Non-sense codons are responsible for terminating peptide chain.

Statement B : Non-sense codons are not recognized by tRNA.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

BIOLOGYLIVE

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133. The base paired structure of DNA implies that it

(1) Can be used to make more copies of DNA

(2) Used as a template to make RNA

(3) Is the hereditary material

(4) 1 and 2

BIOLOGYLIVE

134. Assertion : Regulation of lac operon by repressor is referred to as negative regulation.

Reason : The repressor protein binds to the operator region of the operon.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect

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135. Statement A : The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952).

Statement B : They worked with viruses that infect bacteria called bacteriophages.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

136. In structural gene, the template DNA strand has nucleotide sequences 3'-ATGCATGCATGCATGC- 5. Find the correct and complimentary nucleotide sequence on coding strand.

(1) 5'- ATGCATGCATGCATGC - 3'

(2) 3' - GCATGCATGCATGCAT-5'

(3) 5'- TACGTACGTACGTACG-3'

(4) 3'- TACGTACGTACGTACG-5'

137. Watson and Crick were awarded Nobel prize for showing that

(1) DNA is composed of pentose sugar, nitrogenous base and phosphate

(2) RNA is double helical

(3) DNA is genetic material

(4) DNA is double helical

Title: 200 MCQs Genetics

138. Phosphodiester bond of a segment of ds DNA having 1000 b.p will have?

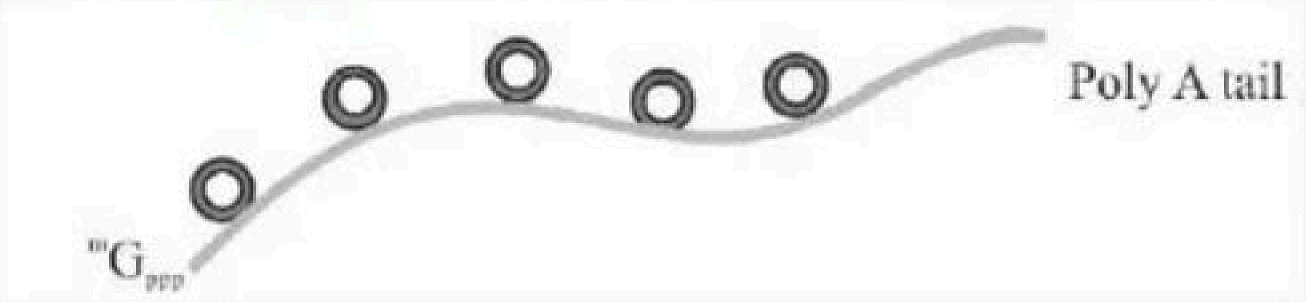
(1) 1000

(2) 2000

(3) 1998

(4) 3000

139. Identify the process showed in the process diagram



(1) Capping

(2) RNA splicing

(3) Poly adenylation

(4) All of these

Title: 200 MCQs Genetics

140. Statement A : The functions are unknown for over 60 per cent of the discovered genes.

Statement B : The human genome contains 3164.7 million nucleotide bases.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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141. DNA from every tissue (such as blood, hairfollicle, skin, bone, saliva, sperm etc.), from an individual show the____

(1) Same degree of polymorphism

(2) No degree of polymorphism

(3) Variable degree of polymorphism

(4) Same degree of monomorphism

BIOLOGYLIVE

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142. Assertion : In Griffith's experiment, a mixture of heat killed virulent bacteria and live non-virulent bacteria S, lead to the death of mice.

Reason : 'Transforming principle' got transferred from heat-killed S strain to R strain and made it virulent.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect

BIOLOGYLIVE

unacademy

143. How many spirals (twins or helices) a DNA of 2000 base pairs will have ?

(1) 200

(2) 2000

(3) 4000

(4) 45.5

BIOLOGYLIVE

unacademy

144. The strand which do not code for anything is called

(1) Coding strand

(2) Non-coding strand

(3) Template strand

(4) Nonsense strand

BIOLOGYLIVE

145. Taylor conducted his experiments to prove semiconservative DNA or chromosome replication on:

(1) E.coli

(2) Vicia faba

(3) Pisum sativum

(4) Bacteriophage

146. The process of transfer of genetic information from DNA to RNA/formation of RNA from DNA is

(1) Transversion

(2) Transcription

(3) Translation

(4) Translocation

BIOLOGYLIVE

unacademy

147. Strand of DNA that functions as template for mRNA transcription is .

i. Coding strand **ii. Non-coding strand**

iii. Sense strand **iv. Antisense strand**

The correct answer is

(1) i and iii

(2) ii and iii

(3) ii and i

(4) ii and iv

BIOLOGYLIVE

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148. Structural genes in Eukaryotes possess coding and non-coding sequences called as 1 and 2 respectively

(1) 1- exons, 2- neutrons

(2) 1- exons, 2- Introns

(3) 1- Introns, 2- Exons

(4) 1-Enhancer, 2- Silencer

BIOLOGYLIVE

149. In actual structure, the tRNA is a compact molecule which looks like_____.

(1) Inverted J

(2) Inverted T

(3) L

(4) Inverted L

Title: 200 MCQs Genetics

150. Read the following statements and find out the incorrect statement.

I. The length of DNA is usually defined as number of nucleotides or base pairs present in it.

II. In DNA, every nucleotide residue has an addition - OH group present at 2' position in the deoxyribose.

III. Nuclein was discovered and named by in Meischer 1969.

IV. Adenine pair with Thymine and Guanine with Cytosine through H-bonds. This makes one strand complementary to the other.

v. The position of double helical model of DNA was also based on the observation of Erwin Charagaff.

(1) II, III and V
(2) II and III only
(3) II, III and IV
(4) III and V only

Title: 200 MCQs Genetics

151. Statement A : Amphibians evolved into reptiles.

Statement B : Fish with stout and strong fins could move on land and go back to water. This was about 350 million years ago.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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152. According to modern synthesis theory, the unit of evolution

(1) Genus

(2) Population

(3) Individual

(4) Species

BIOLOGYLIVE

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153. Which of the following is an example for link species

(1) Lobed finfish

(2) Sea weed

(3) Dodo bird

(4) Chimpanzee

BIOLOGYLIVE

unacademy

154. Embryological support for evolution was disapproved by:

(1) Alfred Wallace

(2) Charles Darwin

(3) Oparin

(4) Karl Ernst von Baer

BIOLOGYLIVE

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155. From his experimints, S.L. Miller produced amino acids by mixing the following in a closed flask:

(1) CH_3, H_2, NH_4 and water vapor at $800^{\circ}C$

(2) CH_4, H_2, NH_3 and water vapor at $600^{\circ}C$

(3) CH_3, H_2, NH_3 and water vapor at $600^{\circ}C$

(4) CH_4, H_2, NH_3 and water vapor at $800^{\circ}C$.

BIOLOGYLIVE

156. Match Column I with Column II:

Column I		Column II	
A	Mesozoic Era	P	Lower invertebrates
B	Proterozoic Era	Q	Fish & Amphibias
C	Cenozoic Era	R	Birds & Reptiles
D	Paleozoic Era	S	Mammals

(1) $\mathrm{A-Q, B-P, C-R, D-S}$

(2) $\mathrm{A-R, B-P, C-Q, D-S}$

(3) $\mathrm{A-P, B-Q, C-S, D-R}$

(4) $\mathrm{A-R, B-P, C-S, D-Q}$

157. In Hardy-Weinberg equation, the frequency of heterozygous individual is represented by :-

(1) pq
(2) 2pq
(3) p^2
(4) q^2

BIOLOGYLIVE

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158. Following are the two statements regarding the origin of life :

i. The earliest organisms that appeared on the earth were non-green and presumably anaerobes.

ii. The first autotrophic organisms were the chemoautotroph's has never released oxygen. Of the above statements which one of the following options is correct?

(1) ii is correct but is false.

(2) Both i and ii are correct.

(3) Both i and ii are false.

(4) i is correct but ii is false.

BIOLOGYLIVE

159. Which of the following had the smallest brain capacity?

(1) Homo sapiens

(2) Homo neanderthalensis

(3) Homo habilis

(4) Homo erectus

160. Flippers of Penguins and Dolphins are examples of :

(1) Convergent evolution

(2) Industrial melanism

(3) Natural selection

(4) Adaptive radiation

BIOLOGYLIVE

unacademy

161. The chronological order of human evolution from early to the recent is :-

(1) Ramapithecus → Homo habilis → Australopithecus → Homo erectus

(2) Australopithecus → Homo habilis → Ramapithecus → Homo erectus

(3) Australopithecus → Ramapithecus → Homo habilis → Homo erectus

(4) Ramapithecus → Australopithecus → Homo habilis → Homo erectus

BIOLOGYLIVE

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162. Natural selection where more individuals acquire specific character value other than the mean character value, leads to:

(1) Directional change

(2) Disruptive change

(3) Random change

(4) Stabilising change

BIOLOGYLIVE

163. Select the correct group/set of Australian Marsupials exhibiting adaptive radiation.

(1) Tasmanian wolf, Bobcat, Marsupial mole

(2) Numbat, Spotted cuscus, Flying phalanger

(3) Mole, Flying squirrel, Tasmanian tiger cat

(4) Lemur, Anteater, Wolf

164. In a population of 1000 individuals 360 belong to genotype AA, 480 to Aa and the remaining 160 to aa. Based on this data, the frequency of allele A in the population is:

(1) 0.4

(2) 0.5

(3) 0.6

(4) 0.7

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165. The extinct human who lived 1,00,000 to 40,000 years ago in Europe, Asia and parts of Africa with short stature, heavy eye browridges, retreating for heads, large jaws with heavy teeth, stocky bodies, a lumbering gait and stooped posture (humpy)

(1) Ramapithecus

(2) Homo habilis

(3) Neanderthal human (Homosapiens neanderthalensis)

(4) Cro-magnan humans

BIOLOGYLIVE

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166. The flippers of the Penguins and Dolphins are the example of the

(1) Adaptive radiation

(2) Natural selection

(3) Convergent evolution

(4) Divergent evolution

BIOLOGYLIVE

unacademy

167. Which of the following statements is not true?

(1) Sweet potato and potato is an example of analogy

(2) Homology indicates common ancestry

(3) Flippers of penguins and dolphins are a pair of homologous organs

(4) Analogous structures are a result of convergent evolution

BIOLOGYLIVE

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168. A gene locus has two alleles , a. If the frequency of dominant allele is 0.4 , then what will be the frequency of homozygous dominant, heterozygous and homozygous recessive individuals in the population?

(1) $0.36(AA); 0.48(Aa); 0.16(aa)$

(2) $0.16(AA); 0.24(Aa); 0.36(aa)$

(3) $0.16(AA); 0.48(Aa); 0.36(aa)$

(4) $0.16(AA); 0.36(Aa); 0.48(aa)$

BIOLOGYLIVE

Title: 200 MCQs Genetics

169. Which of the following refer to correct example(s) of organisms which have evolved due to changes in environment brought about by anthropogenic action?

(a) Darwin's Finches of Galapagos islands

(b) Herbicide resistant weeds

(c) Drug resistant eukaryotes

(d) Man-created breeds of domesticated animals like dogs

(1) (a) and (c) (2) (b), (c) and (d)

(3) only (d) (4) only (a)

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170. Variation in gene frequencies within populations can occur by chance rather than by natural selection. This is referred to as :

(1) Genetic flow

(2) Genetic drift

(3) Random mating

(4) Genetic load

BIOLOGYLIVE

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171. Among the following sets of examples for divergent evolution, select the incorrect option:

(1) Brain of bat, man and cheetah

(2) Heart of bat, man and cheetah

(3) Forelimbs of man, bat and cheetah

(4) Eye of octopus, bat and man

BIOLOGYLIVE

172. Match Column - I with Column - II.

Column I		Column II	
A	Adaptive radiation	P	Selection of resonance varities due to excessive use of herbicides and pesticides
B	Convergent evolution	Q	Bones offorelimbs in man and whale
C	Divergent evolution	R	Wings of butterfly and bird
D	Evolution by anthropogenic action	S	Darwin finches

(1) A - S; B - R; C - Q; D – P

(2) A - R; B - Q; C - P; D - S

(3) A - Q; B - P; C - S; D – R

(4) A - P; B - S; C - R; D - Q

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173. According to Hugo de Vries, the mechanism of evolution is

(1) Phenotypic variations

(2) Saltation

(3) Multiple step mutations

(4) Minor mutations

BIOLOGYLIVE

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174. Analogous structures are a result of

(1) Convergent evolution

(2) Shared ancestry

(3) Stabilizing selection

(4) Divergent evolution

BIOLOGYLIVE

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175. The finches of Galapagos islands provide an evidence

(1) Special creation

(2) Evolution due to mutation

(3) Retrogressive evolution

(4) Biogeographical evolution

BIOLOGYLIVE

176. The process by which organisms with different evolutionary history evolve similar phenotypic adaptations in response to a common environmental challenge is called :

(1) Natural selection

(2) Convergent evolution

(3) Non-random evolution

(4) Adaptive radiation

BIOLOGYLIVE

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177. The similarity of bone structure in the forelimbs of many vertebrates is an example of

(1) Convergent evolution

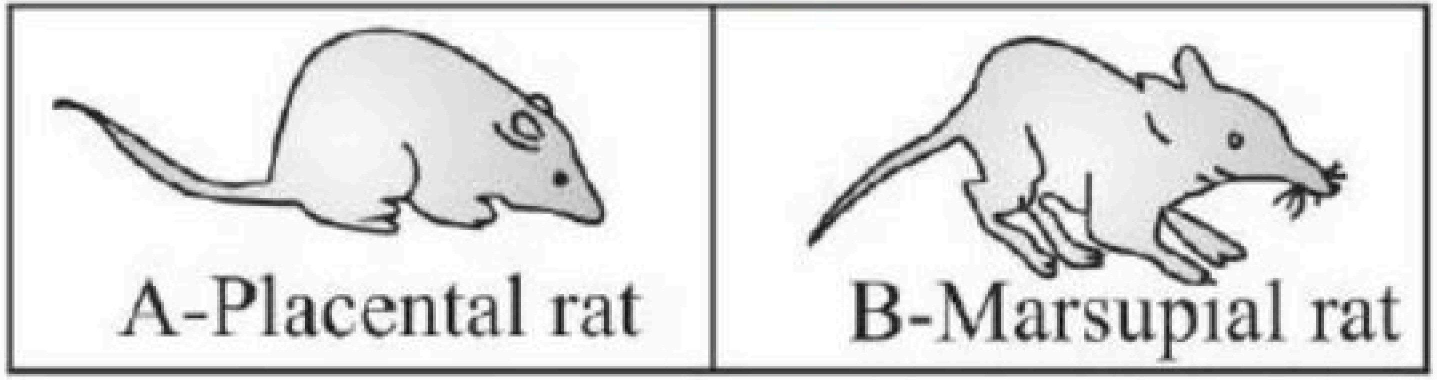
(2) Analogy

(3) Homology

(4) Adaptive radiation

BIOLOGYLIVE

178. The organism A and B are example of:



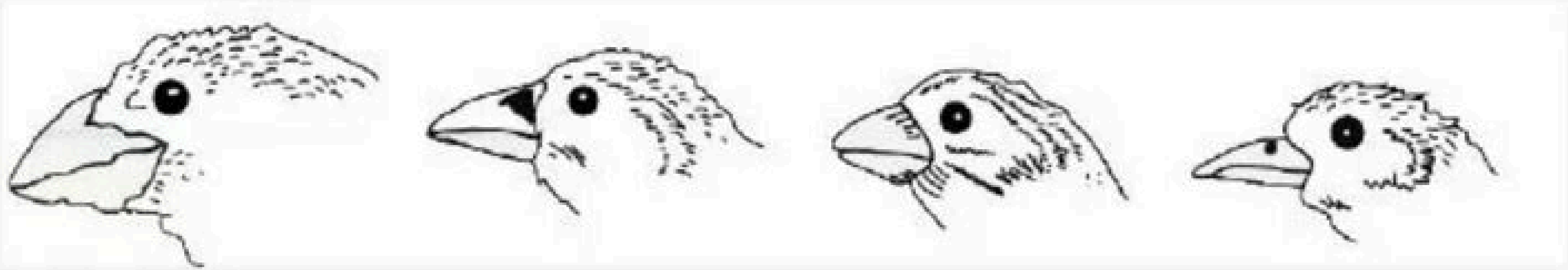
(1) Convergent evolution

(2) Divergent evolution

(3) Adaptive convergence

(4) Both (1) and (3)

179. There is a difference in one morphological character in the birds. The part involved in the change is



(1) Size of eye

(2) Shape of beak

(3) Size of bird size

(4) Size of wings.

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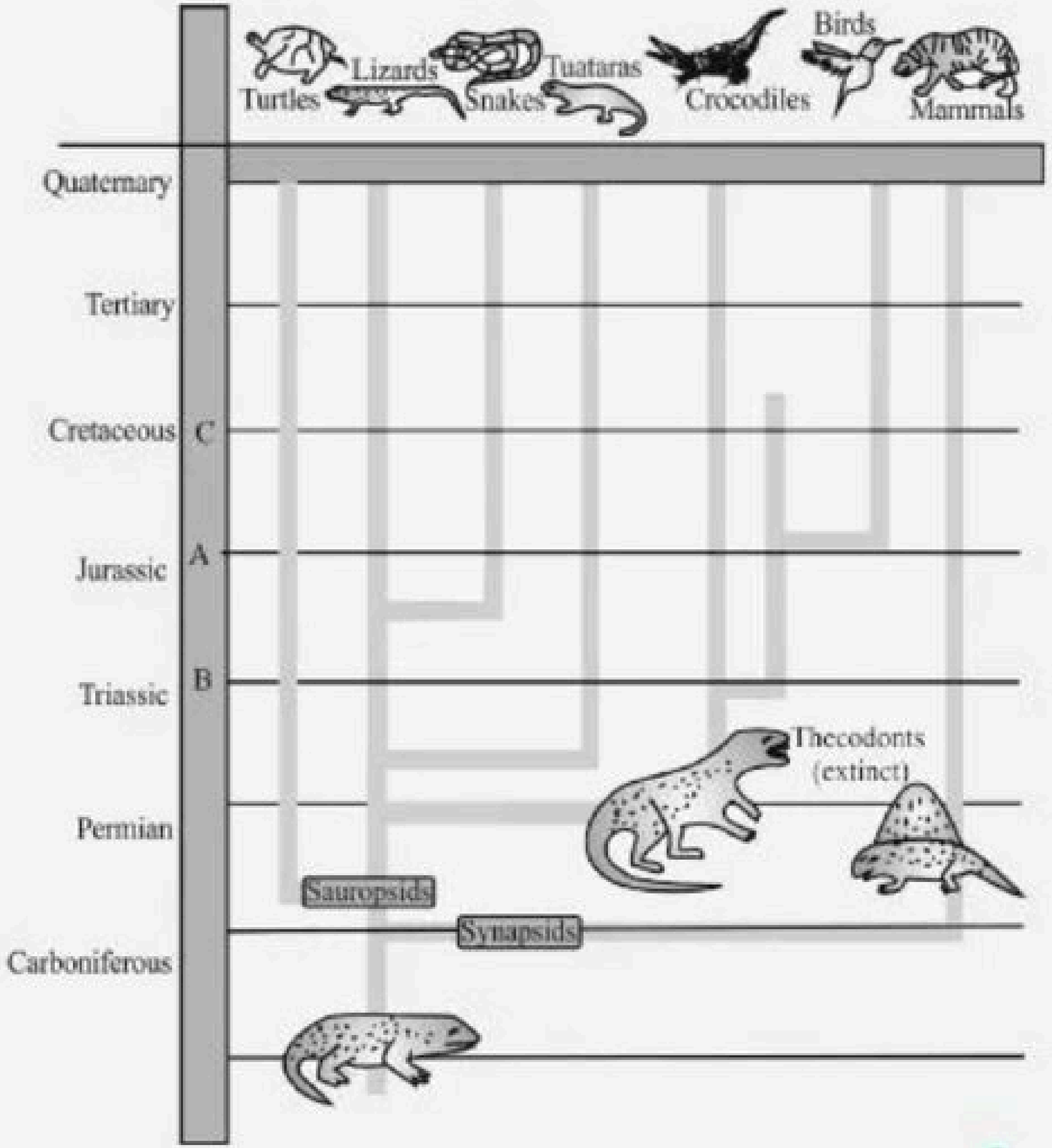
180. Identify the missing time period.

(1) A - 150, B - 100, C -200

(2) A - 150, B - 200, C -100

(3) A - 300, B - 350, C -250

(4) A - 250, B - 350, C -300



BIOLOGYLIVE

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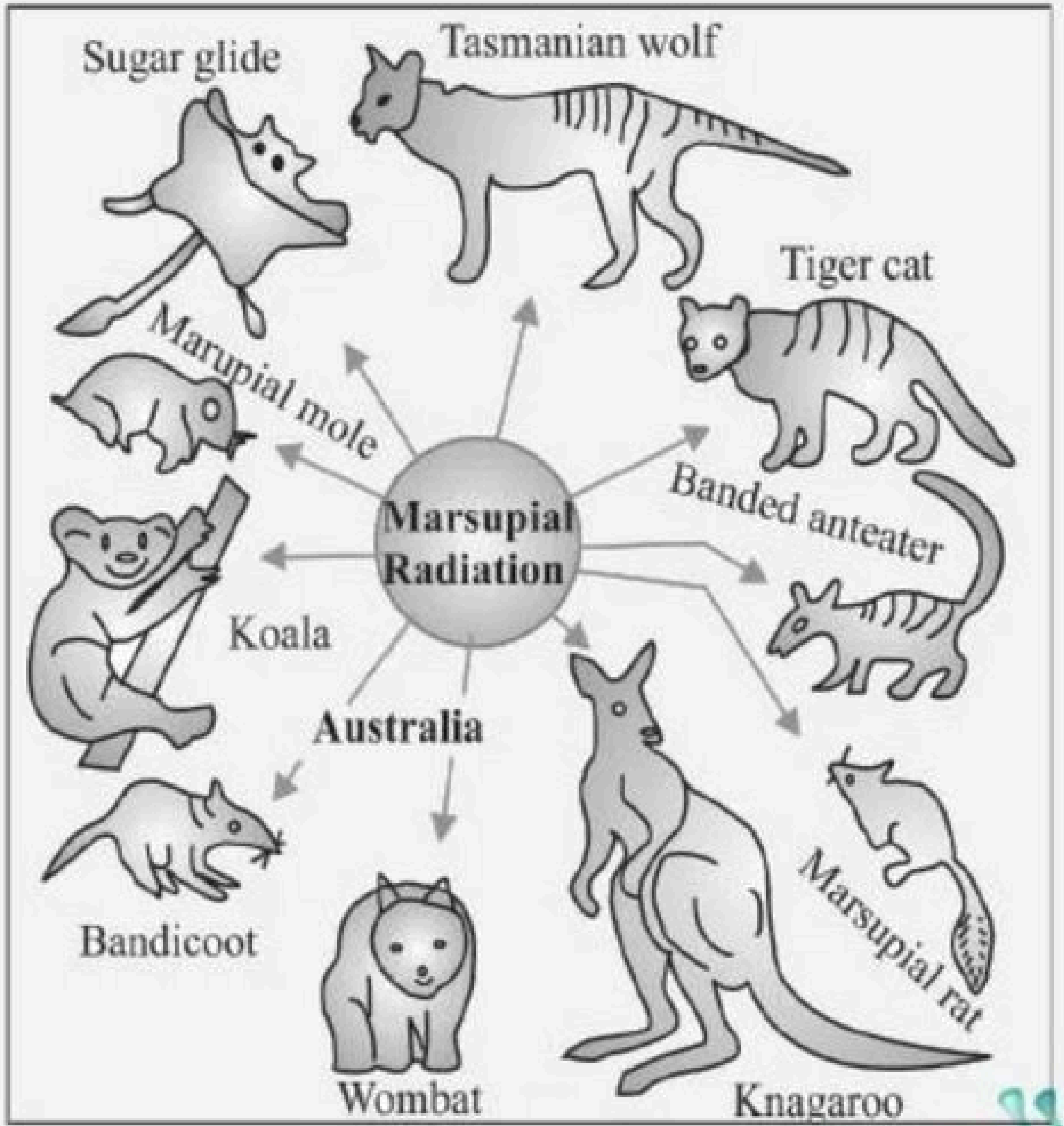
181. The given figure represents

(1) Adaptive Radiation of marsupials of Australia

(2) Convergent evolution of Homo erectus

(3) Divergent evolution of Marupials of Australia

(4) Family tree of Homo erectus



BIOLOGYLIVE

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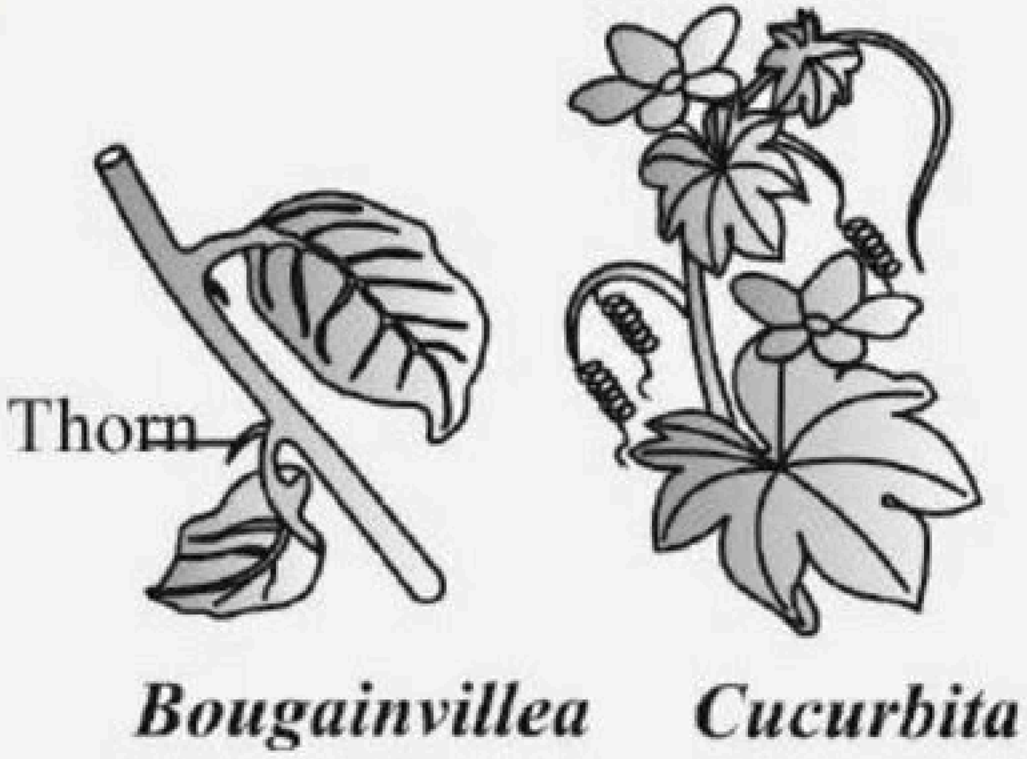
182. The following diagram represents homologous organ, which part of the plant represents homology?

(1) Root tendrils

(2) Leaf tendrils

(3) Stem tendrils

(4) Spine tendrils



BIOLOGYLIVE

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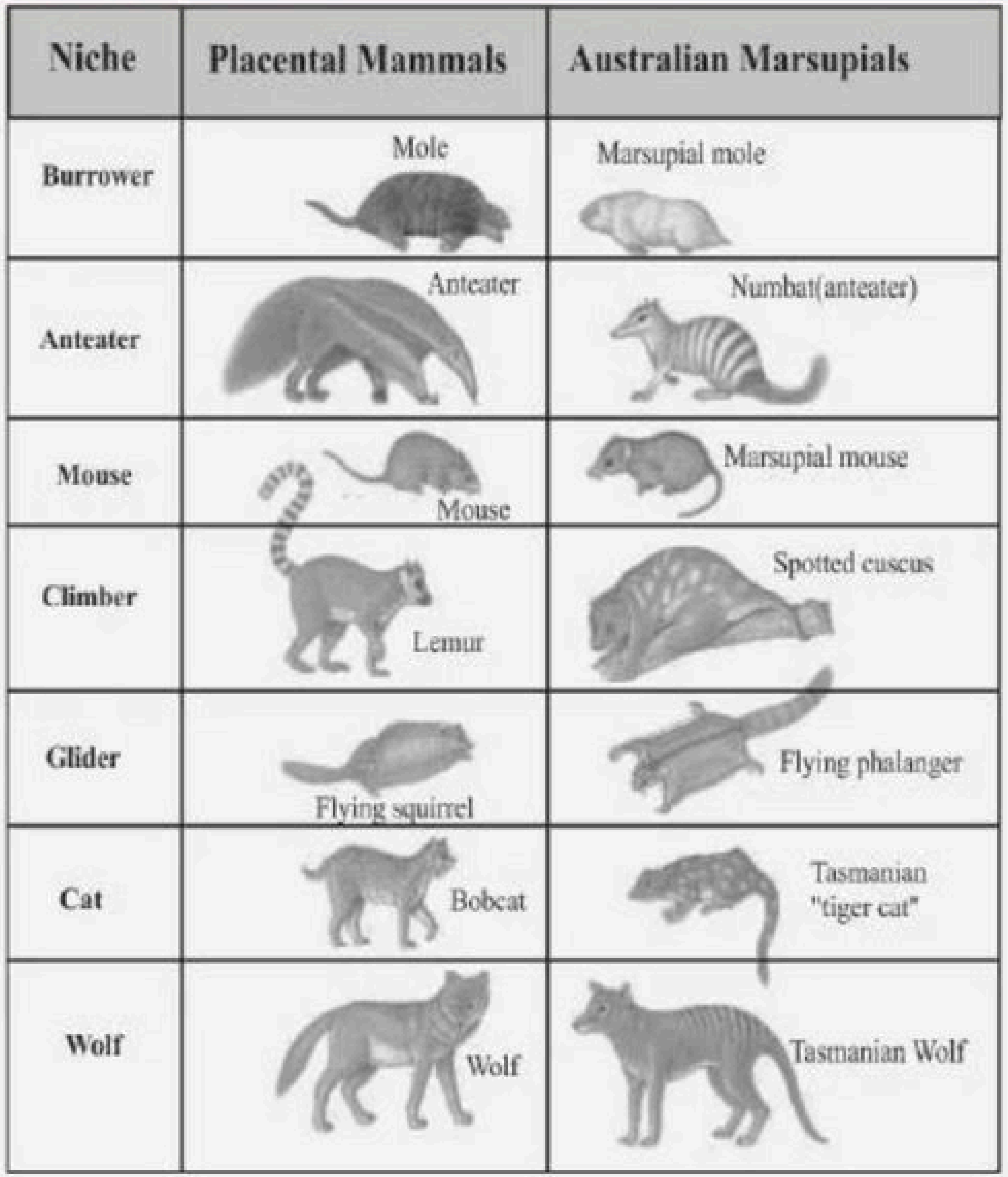
183. The given figure represents

(1) Divergent evolution

(2) Adaptive radiation

(3) Convergent evolution

(4) Classification



Niche	Placental Mammals	Australian Marsupials
Burrower	Mole	Marsupial mole
Anteater	Anteater	Numbat(anteater)
Mouse	Mouse	Marsupial mouse
Climber	Lemur	Spotted cuscus
Glider	Flying squirrel	Flying phalanger
Cat	Bobcat	Tasmanian "tiger cat"
Wolf	Wolf	Tasmanian Wolf

BIOLOGYLIVE

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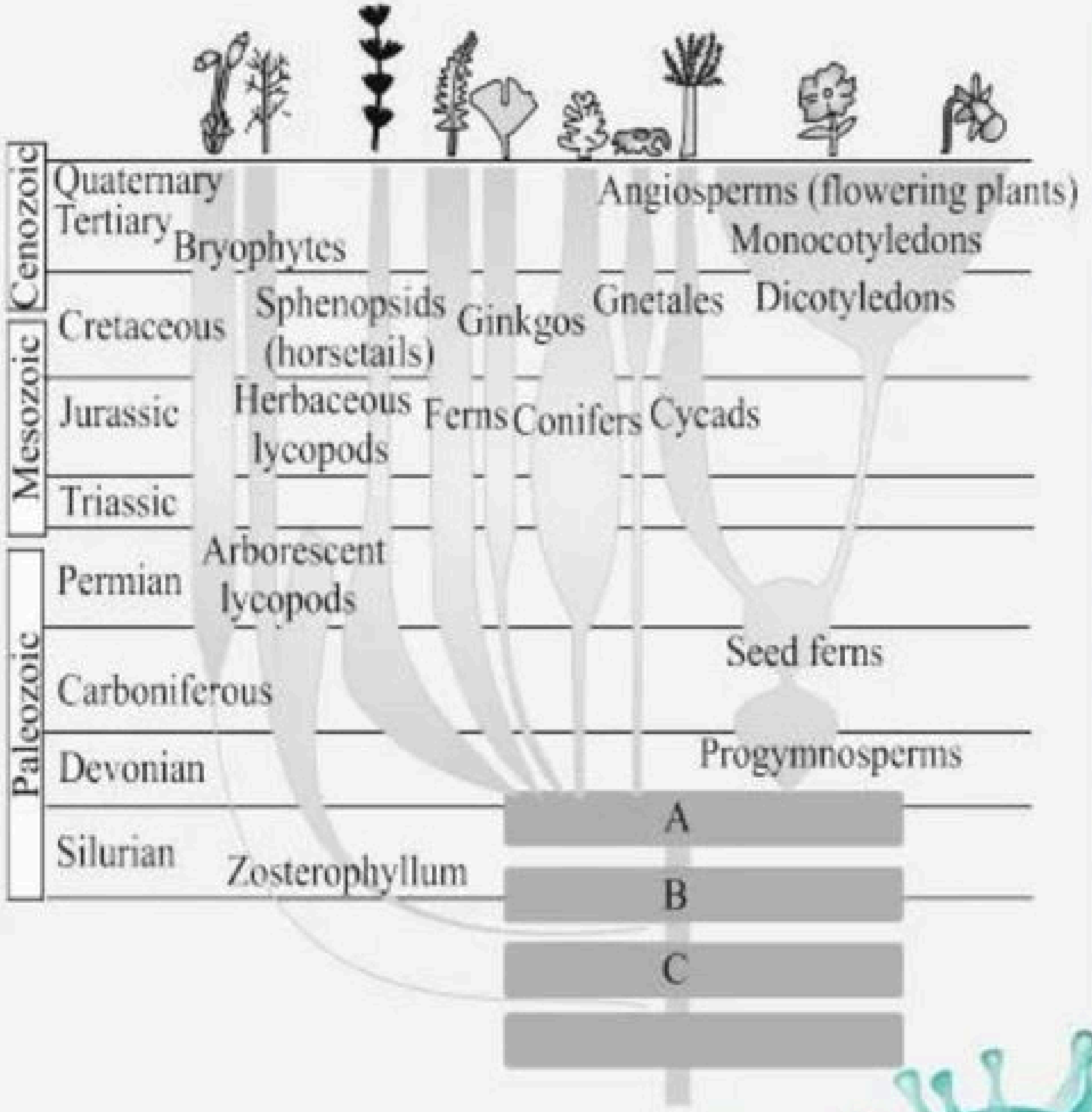
184. Identify the missing ancestors.

(1) A - Chlorophyte ancestor,B - Rhynia type,C - Tracheophyte

(2) A - Psilophyton,B - Rhynia type,C - Tracheophyte

(3) A - Psilophyton,B - Tracheophyte,C - Chlorophyte

(4) A - Psilophyton,B - Rhynia,C - Chlorophyte



BIOLOGYLIVE

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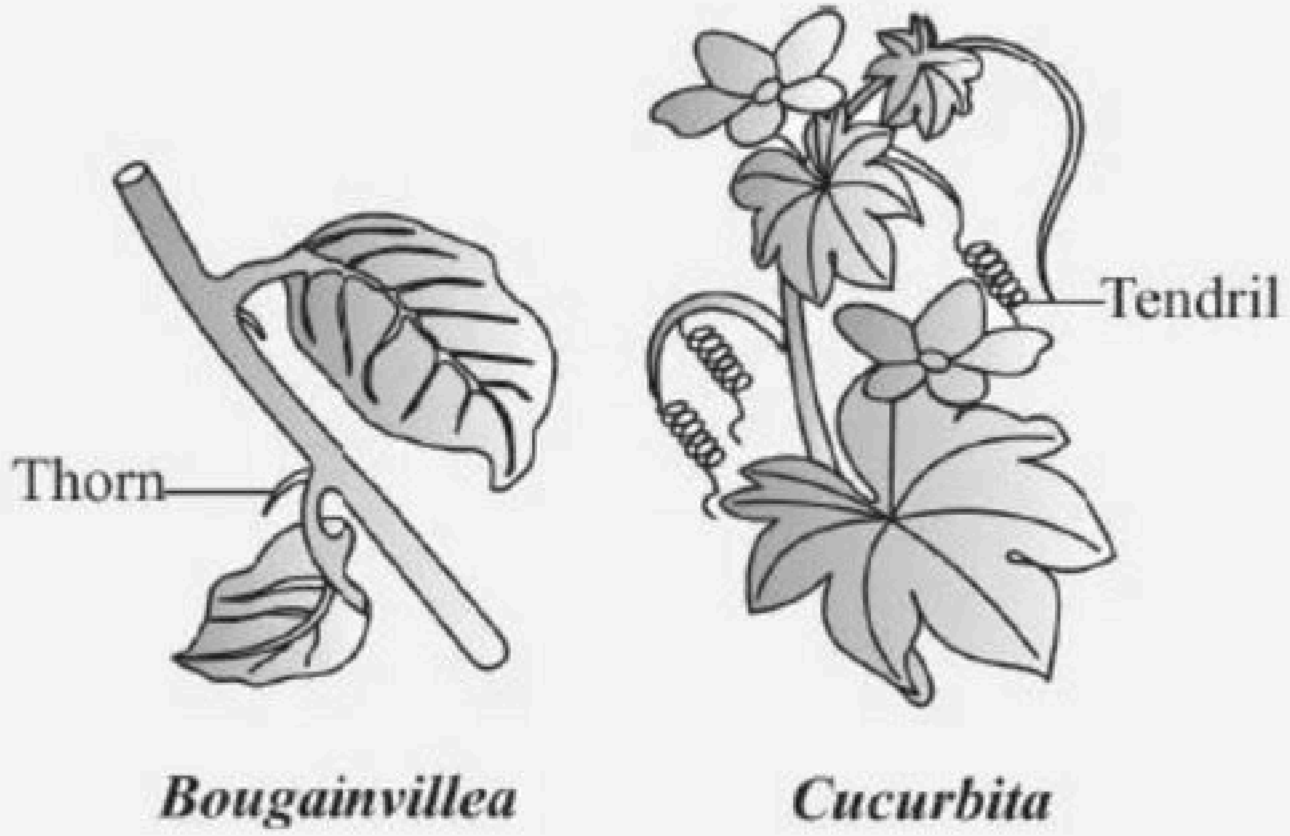
185. The given figure shows an example of

(1) Homologous organs

(2) Convergent evolution

(3) Analogous organs

(4) Vestigial organs



BIOLOGYLIVE

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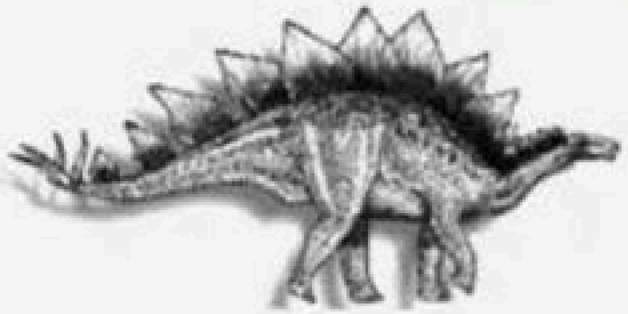
186. Identify the following dinosaur.



(1) Branchiosaurus

(2) Stegosaurus

(3) Archeopteryx

(4) Triceratops

BIOLOGYLIVE

187. Match the following columns.

Column I		Column II	
A	Darwin	P	Branching descent
B	Lamarck	Q	Inheritance of aquired character
C	Pasteur	R	Disapproved spontaneous theory
D	De Vries	S	Mutational theory of inheritance

(1) A – R, B – P, C – Q, D – S
(2) A – P, B – Q, C – R, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

188. Match the following

Column I		Column II	
A	First to use fire	P	*Homo erectus*
B	Ape like primate	Q	*Dryopithecus*
C	Ancestor of modern apes	R	*Australopithecus*
D	Connecting link between ape and man	S	*Propliopethecus*

(1) A – P, B – S, C – Q, D – R

(2) A – Q, B – R, C – P, D – S

(3) A – Q, B – S, C – R, D – P

(4) A – R, B – P, C – S, D – Q

189. Identify the correct match from column-I and column- II:

Column I		Column II	
A	Origin of universe	P	3 bya
B	Origin of Earth	Q	20 bya
C	Origin of first cellular form of life	R	4.5 bya
D	Origin of first noncellular form of life	S	2 bya

(1) A - P, B - Q, C - R, D – S

(2) A - Q, B - R, C - S, D - P

(3) A - Q, B - S, C - R, D – P

(4) A - P, B - R, C - Q, D - S

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190. Match the following columns.

Column I		Column II	
A	Wallace	P	Archipelago
B	Malthus	Q	Influenced Darwin
C	Hardy-Weinberg Law	R	England
D	Industrial melanism	S	$p^2+2pq+q^2=1$

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – P, B – Q, C – S, D – R

BIOLOGYLIVE

Title: 200 MCQs Genetics

191. Match the following columns.

Column I		Column II	
A	Genetic drift	P	Change in the population's allele frequency due to chance alone
B	Natural selection	Q	Selection of better traits
C	Gene flow	R	Immigation or emigration changes the allele frequency
D	Mutation	S	Saltation

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – Q, C – R, D – S

(4) A – Q, B – P, C – R, D – S

192. Match the scientists listed under column ' I ' with ideas

Column I		Column II	
A	Darwin	P	Abiogenesis
B	Oparin	Q	Use and disuse of organs
C	Lamarck	R	Germ plasm theory and neodarwinism
D	August Weisman	S	Evolution by natural selection

(1) A - P, B - S, C - Q, D – R

(2) A - S, B - P, C - Q, D - R

(3) A - Q, B - S, C - R, D – P

(4) A - S, B - R, C - Q, D - P

193. Match the following columns.

Column I		Column II	
A	*Ichthyosaurus*	P	Caught in South Africa in 1938
B	Coelacanth	Q	Fell to form coal deposits
C	Giant pteridophytes	R	Disappeared about 65 mya
D	Dinosaurs	S	Fish-like reptiles in 200 mya

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – S, B – P, C – Q, D – R

194. Match the columns I and II.

Column I		Column II	
A	Oparin	P	America
B	J.B.S. Haldane	Q	Russia
C	Miller	R	England

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

195. Match the following columns.

Column I		Column II	
A	Stabilisation	P	Industrial melanism
B	Directional selection	Q	More individuals possess heterozygous character
C	Disruption selection	R	More individuals possess homozygous character

(1) A – R, B – P, C – Q
(2) A – Q, B – P, C – R
(3) A – P, B – R, C – Q
(4) A – Q, B – R, C – P

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196. Match the columns I and II.

Column I		Column II	
A	*Homo habilis*	P	900 cc
B	*Homo neanderthalensis*	Q	650-800 cc
C	*Homo erectus*	R	1400 cc
D	*Homo sapiens*	S	1350 cc

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – R, C – P, D – S

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197. Match the column I with Column II

Column I		Column II	
A	Mesozoic	P	First amphibians
B	Devonian	Q	300-250 mya
C	Palacocene	R	160 million years
D	Permian	S	Radiation of primitive mammals

(1) A – R, B – P, C – R, D – Q

(2) A – R, B – P, C – S, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

BIOLOGYLIVE

198. Select the correct statements from the following.

I. Dryopithecus and Ramapithecus were existing about 15 mya.

II. They were hairy and walked like gorillas and chimpanzees.

III. Ramapithecus was more ape-like.

IV. Dryopithecus was more man-like.

(1) I and II

(2) III and IV

(3) II and III

(4) I, II, III and IV

199. Read the following statements and select the correct option.

I. Branching descent and natural selection are the two key concepts of Darwinian Theory of Evolution

II. When more individuals of a population acquire a mean character value, it is called disruption.

III. Hardy-Weinberg equilibrium says that allele frequencies in a population are stable and is constant from generation to generation.

IV. Genetic drift changes the existing gene or allelic frequency in future generations

(1) III alone is correct

(2) II, III and IV are correct

(3) I, III and IV are correct

(4) Both I and III are correct

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200. Statement A : Whales and mammals share similarities in the pattern of bones of forelimbs.

Statement B : These organisms suggest occurrence of Convergent evolution.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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THANK YOU

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